

PRAKTIKUM SISTEM OPERASI

MODUL X

SHELL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Modifikasi Syscall chname.

Jawaban:



```
/* in file addargs.c */
extern status addargs(pid32, int32, int32[], char *, void *);

/* in file arp.c */
extern void arp_init(void);
extern status arp_resolve(uint32, byte[]);
extern void arp_in(struct arppacket *);
extern int32 arp_alloc(void);
extern void arp_ntoh(struct arppacket *);
extern void arp_hon(struct arppacket *);

/* in file ascdade.c */
extern status ascdade(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
```

a. Edit prototypes.h.

1. gedit ~/xinu_m10/include/prototypes.h
- 2.

Cari baris chname, ubah menjadi:

1. /* in file chname.c */
2. extern pri16 chname(pid32, char *, uint32);
- 3.

Save.

```
#include <xinu.h>

pri16 chname(
    pid32 pid,
    char *newname,
    uint32 namelen
) {
    intmask mask;
    struct procent *prptr;

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16) SYSERR;
    }

    prptr = &proctab[pid];
    strncpy(prptr->prname, newname, namelen);
    prptr->prname[namelen] = '\0';

    restore(mask);
    return OK;
}
```

C Tab Width: 8 Ln 19, Col 33 INS

b. Edit chname.c

1. gedit ~/xinu_m10/system/chname.c
- 2.

Ubah seluruh isi fungsi menjadi:

```
1. #include <xinu.h>
2.
3. pri16 chname(
4.     pid32 pid,          /* ID of process to change */
5.     char *newname,
6.     uint32 namelen
7. ) {
8.     intmask mask;
9.     struct procent *prptr;
10.
11.     mask = disable();
12.     if (isbadpid(pid)) {
13.         restore(mask);
14.         return (pri16) SYSERR;
15.     }
16.
17.     prptr = &proctab[pid];
18.     strncpy(prptr->prname, newname, namelen);
19.     prptr->prname[namelen] = '\0';
```

```
20.  
21.     restore(mask);  
22.     return OK;  
23. }  
24.
```

Save.

2. Membuat Perintah namecmd.

Jawaban:



The screenshot shows a gedit text editor window titled 'xsh_namecmd.c' with the path '~/xinu_m10/shell'. The window contains the following C code:

```
/* Nama: Marvel Sanjaya Setiawan NIM:231104053 */  
#include <xinu.h>  
  
shellcmd xsh_namecmd(int32 nargs, char *args[]) {  
    printf("My new command");  
    chname(2, "hello", 5);  
    return 0;  
}
```

The status bar at the bottom indicates 'C', 'Tab Width: 8', 'Ln 6, Col 17', and 'INS'.

```
*/
syscall chprio(
    pid32    pid,          /* ID of process to change */
    pri16    newprio       /* New priority */
)
{
    intmask mask;          /* Saved interrupt mask */
    struct proctab *prptr; /* Ptr to process's table entry */
    pri16 oldprio;         /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

a. Buat file xsh_namecmd.c

1. gedit ~/xinu_m10/shell/xsh_namecmd.c
- 2.

Isi dengan:

```
1. /* Nama: XXX NIM: XXX */
2. #include <xinu.h>
3.
4. shellcmd xsh_namecmd(int32 nargs, char *args[]) {
5.     printf("My new command");
6.     // pid 2 = netin; mengubah nama proses pid 2 yaitu netin menjadi hello
7.     chname(2, "hello", 5);
8.     return 0;
9. }
10.
```

Save.

```
/* in file xsh_ps.c */
extern shellcmd xsh_ps      (int32, char *[]);

/* in file xsh_sleep.c */
extern shellcmd xsh_sleep   (int32, char *[]);

/* in file xsh_udpdump.c */
extern shellcmd xsh_udpdump (int32, char *[]);

/* in file xsh_udpecho.c */
extern shellcmd xsh_udpecho (int32, char *[]);

/* in file xsh_udpserver.c */
extern shellcmd xsh_udpserver (int32, char *[]);

/* in file xsh_uptime.c */
extern shellcmd xsh_uptime   (int32, char *[]);

/* in file xsh_help.c */
extern shellcmd xsh_help     (int32, char *[]);
extern shellcmd xsh_mycmd(int32, char *[]);

/* in file xsh_tee.c */
extern shellcmd xsh_tee      (int32, char *[]);

extern shellcmd xsh_mycmd    (int32, char *[]);

extern shellcmd xsh_namecmd  (int32, char *[]);
```

b. Daftarkan ke shprototypes.h

1. gedit ~/xinu_m10/include/shprototypes.h
- 2.

Tambahkan:

1. extern shellcmd xsh_namecmd(int32, char *[]);
- 2.

Save.

```
{ "arpecho", xsh_arpecho},
{ "arp", xsh_arp},
{ "cat", xsh_cat},
{ "clear", xsh_clear},
{ "date", xsh_date},
{ "devdump", xsh_devdump},
{ "echo", xsh_echo},
{ "help", xsh_help},
{ "ls", xsh_ls},
{ "kill", xsh_kill},
{ "memdump", xsh_memdump},
{ "memstat", xsh_memstat},
{ "ns", xsh_ns},
{ "netinfo", xsh_netinfo},
{ "ping", xsh_ping},
{ "ps", xsh_ps},
{ "sleep", xsh_sleep},
{ "tee", xsh_tee},
{ "udp", xsh_udpdump},
{ "udpecho", xsh_udpecho},
{ "udpserver", xsh_udpserver},
{ "uptime", xsh_uptime},
{ "mycmd", xsh_mycmd},
{ "mycmd", xsh_mycmd},
{ "?", xsh_help},
{ "namecmd", xsh_namecmd},
};
```

c. Daftarkan ke shell.c

1. gedit ~/xinu_m10/shell/shell.c
- 2.

Cari tabel command, tambahkan:

1. { "namecmd", xsh_namecmd},
- 2.

Save.

1. /* chprio.c - chprio */
2. #include <xinu.h>
- 3.
4. syscall chprio(
5. pid32 pid,
6. pri16 newprio
7.)
8. {
9. intmask mask;
10. struct procent *prptr;
11. pri16 oldprio;
- 12.
13. mask = disable();

```

14.     if (isbadpid(pid)) {
15.         restore(mask);
16.         return SYSERR;
17.     }
18.     if (newprio <= 0) {
19.         restore(mask);
20.         return SYSERR;
21.     }
22.     prptr = &proctab[pid];
23.     oldprio = prptr->prprio;
24.     prptr->prprio = newprio;
25.     restore(mask);
26.     return 0xffff & oldprio;
27. }
28.

```

d. Compile

```

1. cd ~/xinu_m10/compile
2. make clean && make
3.

```

3. Test Hasil

Jawaban:

The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu_m10/compile". The terminal displays the XINU boot sequence, including the "Welcome to Xinu!" message. The user enters the command "ps", and the terminal shows a table of running processes. The table has columns for Pid, Name, State, Prio, Ppid, Stack Base, Stack Ptr, and Stack Size. The processes listed are prnull, ipout, netin, Main process, shell, and ps. The user then enters the command "namecmd", and the terminal shows the output "My new command". The user then enters the command "ps" again, and the terminal shows the same table of running processes. The terminal also shows the command "CTRL-A Z for help" and the status "115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0".

```

-- -- ----- - - ----
-----

Welcome to Xinu!

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull                ready  0    0 0x005FDFFC 0x00FFFFEE4 8192
1 ipout                 wait  500   0 0x005FBFFC 0x005FBF30 8192
2 netin                 wait  500   0 0x005F9FFC 0x005F9ED0 8192
4 Main process          recv  20    3 0x005E7FFC 0x005E7F44 65536
5 shell                  recv  50    4 0x005D7FFC 0x005D79AC 8192
6 ps                     curr  20    5 0x005F7FFC 0x005F7FC4 8192

xsh $ namecmd
My new commandxsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull                ready  0    0 0x005FDFFC 0x00FFFFEE4 8192
1 ipout                 wait  500   0 0x005FBFFC 0x005FBF30 8192
2 hello                 wait  500   0 0x005F9FFC 0x005F9ED0 8192
4 Main process          recv  20    3 0x005E7FFC 0x005E7F44 65536
5 shell                  recv  50    4 0x005D7FFC 0x005D79AC 8192
8 ps                     curr  20    5 0x005F7FFC 0x005F7FC4 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```


a. sudo minicom

b. Di terminal Xinu jalankan berurutan

1. ps
2. namecmd
3. ps
- 4.